

VII · WHAT IS MEASURED

Everything measured on the Voynich Manuscript

The language tests, the stroke runs, the failed Genesis fit, the three readings, and what is still open.

Wallet: `bc1qd8wcrsmkk0ghegf5t2jsezg2ve7rtx6vvc79lx`

Everything measured on the Voynich Manuscript, positive and negative at the same size. A proposed decipherment has to clear all of it.

1

Against ten languages

Key-free measures only — word length, letter entropy, first and last letter fixedness, immediate repeats, types, hapax. The Voynich Manuscript stands 2.7 to 4.4 of the languages' own standard deviations from every one of them, where they sit 0.6 to 2.7 from each other. Nearest is Middle High German; turned or mirrored, nearest is Hebrew at 3.4.

TEXT	LETTER ENTROPY H1	CONDITIONAL H2
Old Czech (Dalimil, Hus)	4.77	3.53
Czech (1613 Bible and later prose)	4.69	3.60
Church Slavonic, Glagolitic (Zographensis)	4.68	3.29

TEXT	LETTER ENTROPY H1	CONDITIONAL H2
Hebrew (Genesis, Exodus, consonantal)	4.23	3.47
Occitan	4.22	3.21
Early New High German	4.18	3.23
Italian	4.12	3.15
Latin	4.05	3.26
Old French	4.03	3.18
Middle High German	4.00	3.04
MS, compound glyphs as one sign	3.87	2.35
MS, EVA as transliterated	3.86	2.12
MS, the same turned (words reversed, order reversed)	3.85	2.29

Corpora fetched from Wikisource: Old Czech (Dalimil, Hus's Dcerka), Hebrew (Genesis, Exodus, consonantal), Church Slavonic in Glagolitic (Codex Zographensis), and the rest.

source artifacts/voynich_2026-09-19/language/structure/structure2.json, structure2.txt

2

The stroke runs stop at four

Runs of plain strokes closed by the hooked stroke, counted with the hook, over the whole book:

RUN	1 STROKE	2	3	4	5
Count	137	1,719	4,131	165	1

They stop at four exactly as a scribe's numerals do — j, ij, iij, iiij. Read as powers of four the sequence runs 4, 16, 64, 256, 253, 241, 193, 1, closing on the order-8 subgroup of the field. Four-stroke runs follow a 121 times and o 41 times, together 162; and one-stroke runs follow a 121 times as well — the same count.

The 121 words with a four-stroke run after a sum to $6,562 = 25 \times 257 + 137$: residue **137 EYE OF HORUS**, quotient **25 = Δ^2** .

source language/structure/four_strokes.txt, minims.txt, baalaab.txt

3

Rare glyphs keep strict company

- b is always last, after e — 15 of 15.
- u is always last or alone — 8 of 8.
- v sits on f57v, 10 of 11.
- x sits on the rosettes, f57v and f66r.
- There is no a followed by u anywhere in the book, against 4,279 instances of a followed by two strokes. Occitan runs 97.9 "au" per 10,000 letters, Early New High German 86.1, Latin 54.8, Old Czech 1.0.

Sparse is not absent. A letter that appears rarely but under a strict rule is data, in the way Q is data in English.

4

The Genesis fit, which failed

NEGATIVE RESULT

Genesis 1–13 was fetched in Hebrew and reduced to 337 verses with their gematria, counts checked. Solving for additive glyph values so that the Voynich Manuscript's entries, lines or words equal the verses or words of Genesis in order gives **zero exact hits**, no better than a shuffled control, and no solution in GF(257).

The fitted values are still data, and read by the field they are: k → 78 GENESIS · d → 256 · u → 129 · a → 228 = $RIT0^{-1}$ · m → 111 · p → 233 · s → 139 · i → 232 · y → 244; weighted sum $\equiv 2$.

```
source stars/genesis_fit.txt, genesis_fit_words.txt,
genesis_fit_values.txt, heb_verses.py
```

5

What the three readings give

Three readings were computed over all 322 starred entries of the Voynich Manuscript: upright, mirrored (each line right to left, lines in order) and half-turned (the whole entry reversed). Letter sums and cells never change; the φ part does.

READING	DATED ENTRY POINTS, YEARS 1–1420	ALL 322 TOGETHER
upright	94	253
mirrored — the reading	80	105
half turn	78	243

The mirror is not chosen because it scores highest on this measure. It is chosen because it is the only one that puts 913 and 203 in order at entry 6.

6

What is still open

1. Lines inside an entry: down only (203), or up as well (192).
2. Ring order on a zodiac page. Every result takes the transcribers' list order; no other order has been tested.
3. Three of the five starless figures do not fall on a new Moon.
4. Capricorn and Aquarius are lost with folio 74, so mid-December to early February has no figure to land on.
5. Whether the faint second line above the long P of f85r1 is real.

6. The centre month-words, and whose hand wrote them.

PROVENANCE

Where every figure comes from

DOMAIN	FILES
the field	emerald_tablet.json — 1,460 words, 257 cells, with its page, plane, reflected and sides files and the Rust crate's tests
the text	transliteration/ZL3b-n.txt — the EVA transliteration, 227 sides, 5,385 lines, 36,278 words, 194,262 letters
the images	Voynich_Manuscript.pdf (209 page images) and full-size Beinecke IIIF spreads for f68r and every zodiac page
the figures	beinecke/figures/ — PROTOCOL.txt, twelve per-page tables, all_299_figures.json, per-page crops
the sky	eclipses/ — NASA catalogues 1301–1500, prague_view.json, moon_pleiades.json; prague_1408/sky/skylib.py
the language	language/structure/ and language/marginalia/corpus/ — ten reference corpora
the record	INVESTIGATION_2026-09-21.md, MANIFEST_2026-09-21.sha256, and the R1 offloads

Every page on this site is generated by `END/build.py`, which reads those files at build time. No number here is typed in by hand.

Built 2026-09-21 from the corpus and `~/Documents/artifacts`. Every figure on this site is read from a file at build time by `build.py`; none is typed in.

Emerald tablet: 1,460 words over 257 cells · field GF(257), $g = 3$ · the derivation this answers is at

start.rito.one.

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